

Algebras

Linear Algebra

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Algebras

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Structure Constants

Division Algebras

The Algebra $\mathbb{H}$

The Generator of an Algebra

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Derivation of an Algebra

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Definition

An **algebra** over $\mathbb{C}$ is a vector space over $\mathbb{C}$, together with a binary operation $\mathcal{A} \times \mathcal{A} \rightarrow \mathcal{A}$ called the **multiplication**. The image of $(\mathbf{a}, \mathbf{b}) \in \mathcal{A}^2$ under this mapping is denoted by $\mathbf{ab}$, and it satisfies the following two relations

$$\mathbf{a}(\beta\mathbf{b} + \gamma\mathbf{c}) = \beta\mathbf{ab} + \gamma\mathbf{ac}$$

$$(\beta\mathbf{b} + \gamma\mathbf{c})\mathbf{a} = \beta\mathbf{ba} + \gamma\mathbf{ca}$$

for all $\mathbf{a}, \mathbf{b}, \mathbf{c} \in \mathcal{A}, \beta, \gamma \in \mathbb{C}$. The dimension of the underlying vector space is called the **dimension of the algebra**.

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Definition

An algebra $\mathcal{A}$ is called **associative** if the product satisfies $\mathbf{a}(\mathbf{bc}) = (\mathbf{ab})\mathbf{c}$ and **commutative** if it satisfies $\mathbf{ab} = \mathbf{ba}$. An algebra with **identity** is an algebra that has an element $\mathbf{1}$ satisfying $\mathbf{1a} = \mathbf{a1} = \mathbf{a}$. An element $\mathbf{b}$ of an algebra with identity is said to be a **left inverse** of $\mathbf{a}$ if $\mathbf{ba} = \mathbf{1}$. **Right inverse** is defined similarly. The identity is also called **unit**, and an algebra with identity is also called a **unital algebra**.

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Remark

Taking $\beta = 1 = -\gamma$ and $\mathbf{b} = \mathbf{c}$ in the definition above leads immediately to

$$\mathbf{a}\mathbf{0} = \mathbf{0}\mathbf{a} = \mathbf{0} \quad \forall \mathbf{a} \in \mathcal{A}.$$

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Proposition 2.1

The identity of an algebra is unique.

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Example 2.2

On the usual three-dimensional Euclidean space $\mathbb{R}^3$ define the product $\mathbf{u} \times \mathbf{v}$ by

$$\mathbf{u} \times \mathbf{v} = \varepsilon_{ijk} u_j v_k \mathbf{e}_i,$$

where $\{\mathbf{e}_i\}_{i=1}^3$ is a basis for $\mathbb{R}^3$ and ε_{ijk} is the **Levi-Civita symbol** defined by

$$\varepsilon_{ijk} = \begin{cases} 1 & \text{if } (i, j, k) \text{ is an even permutation of } (1, 2, 3), \\ -1 & \text{if } (i, j, k) \text{ is an odd permutation of } (1, 2, 3), \\ 0 & \text{otherwise.} \end{cases}$$

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Proposition 2.3

If $\mathcal{A}$ is an associative algebra and $\mathbf{a} \in \mathcal{A}$ has both a left inverse $\mathbf{b}$ and a right inverse $\mathbf{c}$, then the two are equal.

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Theorem 2.4

Let $\mathcal{A}$ be an associative algebra with identity. If $\mathbf{a} \in \mathcal{A}$ has a right and left inverse, then they are equal and this single inverse is unique. We denote it by $\mathbf{a}^{-1}$. If $\mathbf{a}$ and $\mathbf{b}$ are invertible, then $\mathbf{ab}$ is also invertible, and

$$(\mathbf{ab})^{-1} = \mathbf{b}^{-1}\mathbf{a}^{-1}.$$

Definition

Let $\mathcal{A}$ be an algebra and $\mathcal{A}'$ a linear subspace of $\mathcal{A}$. If $\mathcal{A}'$ is closed under multiplication, i.e., if $\mathbf{ab} \in \mathcal{A}'$ whenever $\mathbf{a} \in \mathcal{A}'$ and $\mathbf{b} \in \mathcal{A}'$, then $\mathcal{A}'$ is called a **subalgebra** of $\mathcal{A}$. Let $\mathcal{A}$ be an associative algebra and S a subset of $\mathcal{A}$, then the **subalgebra generated by S** is the collection of all linear combinations of

$$\mathbf{s}_1 \mathbf{s}_2 \dots \mathbf{s}_k, \quad \mathbf{s}_i \in S.$$

If S consists of a single element $\mathbf{s}$, then the subalgebra generated by S is the set of polynomials in S .

Example 2.5

Let $\mathcal{A}$ be an unital algebra, then the linear space

$$\text{Span}\{\mathbf{1}\} = \{\alpha\mathbf{1} \mid \alpha \in \mathbb{C}\}$$

is a subalgebra of $\mathcal{A}$.

Definition

Let $\mathcal{A}$ be an algebra. The set of elements of $\mathcal{A}$ which commute with all elements of $\mathcal{A}$ is called the **centre** of $\mathcal{A}$ and denoted by $\mathcal{Z}(\mathcal{A})$.

Proposition 2.6

$\mathcal{Z}(\mathcal{A})$ is a subspace of $\mathcal{A}$, and if $\mathcal{A}$ is associative, then $\mathcal{Z}(\mathcal{A})$ is a subalgebra of $\mathcal{A}$.

Definition

A unital algebra $\mathcal{A}$ is called **central** if $\mathcal{Z}(\mathcal{A}) = \text{Span}\{\mathbf{1}\}$.

Example 2.7

Consider the algebra $\mathcal{S}$ with basis $\{\mathbf{e}_i\}_{i=0}^3$ and multiplication table given by Figure 1. Show that $\mathcal{S}$ is central.

	$\mathbf{e}_0$	$\mathbf{e}_1$	$\mathbf{e}_2$	$\mathbf{e}_3$
$\mathbf{e}_0$	$\mathbf{e}_0$	$\mathbf{e}_1$	$\mathbf{e}_2$	$\mathbf{e}_3$
$\mathbf{e}_1$	$\mathbf{e}_1$	$\mathbf{e}_0$	$\mathbf{e}_3$	$\mathbf{e}_2$
$\mathbf{e}_2$	$\mathbf{e}_2$	$-\mathbf{e}_3$	$-\mathbf{e}_0$	$\mathbf{e}_1$
$\mathbf{e}_3$	$\mathbf{e}_3$	$-\mathbf{e}_2$	$-\mathbf{e}_1$	$\mathbf{e}_0$

Figure 1: The multiplication table for $\mathcal{S}$

Definition

Let A and B be subsets of an algebra $\mathcal{A}$. We denote by AB the set of elements in $\mathcal{A}$ which can be written as the sum of products of an element in A by an element of B :

$$AB := \left\{ \mathbf{x} \in \mathcal{A} \mid \mathbf{x} = \sum_k \mathbf{a}_k \mathbf{b}_k, \mathbf{a}_k \in A, \mathbf{b}_k \in B \right\}.$$

In particular,

$$\mathcal{A}^2 := \left\{ \mathbf{x} \in \mathcal{A} \mid \mathbf{x} = \sum_k \mathbf{a}_k \mathbf{b}_k, \mathbf{a}_k, \mathbf{b}_k \in \mathcal{A} \right\}$$

is called the **derived algebra** of $\mathcal{A}$.

Definition

Given an algebra $\mathcal{A}$, in which $(\mathbf{a}, \mathbf{b}) \mapsto \mathbf{ab}$, we can obtain a second algebra $\mathcal{A}^{\text{op}}$ in which $(\mathbf{a}, \mathbf{b}) \mapsto \mathbf{ba}$. We write

$$(\mathbf{ab})^{\text{op}} = \mathbf{ba}$$

and call $\mathcal{A}^{\text{op}}$ the algebra **opposite** to $\mathcal{A}$.

Example 2.8

If $\mathbf{T} : \mathcal{V} \rightarrow \mathcal{U}$ and $\mathbf{S} : \mathcal{U} \rightarrow \mathcal{W}$ are linear operators, then the composition $\mathbf{S} \circ \mathbf{T} : \mathcal{V} \rightarrow \mathcal{W}$ is also a linear operator. This product, however, is not defined on a single vector space, but is such that it takes an element in $\text{Hom}(\mathcal{V}, \mathcal{U})$ and another element in a second vector space $\text{Hom}(\mathcal{U}, \mathcal{W})$ to give an element in yet another vector space $\text{Hom}(\mathcal{V}, \mathcal{W})$. An algebra requires a single vector space. We can accomplish this by letting $\mathcal{V} = \mathcal{U} = \mathcal{W}$. Then the three spaces of linear transformations collapse to the single space $\text{End}(\mathcal{V})$, the set of endomorphisms of $\mathcal{V}$, to which $\mathbf{T}, \mathbf{S}, \mathbf{ST} := \mathbf{S} \circ \mathbf{T}$, and $\mathbf{TS} := \mathbf{T} \circ \mathbf{S}$ belong.

Example 2.9

The vector space $C^r(\Omega)$ of complex-valued functions defined on a complex subset Ω , which have derivatives up to order r . The multiplication is defined pointwise: If $f \in C^r(\Omega)$ and $g \in C^r(\Omega)$, then

$$(fg)(z) := f(z)g(z) \quad \forall z \in \Omega.$$

This algebra is commutative and associative, and has the identity element $f(z) = 1$.

Definition

Given an algebra $\mathcal{A}$ and a basis $B = \{\mathbf{e}_i\}_{i=1}^N$ for the underlying vector space, one can write

$$\mathbf{e}_i \mathbf{e}_j = \sum_{k=1}^N c_{ij}^k \mathbf{e}_k, \quad c_{ij}^k \in \mathbb{C}.$$

The complex numbers c_{ij}^k , the components of the vector $\mathbf{e}_i \mathbf{e}_j$ in the basis B , are called the **structure constants** of $\mathcal{A}$.

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Definition

*A unital algebra all of whose nonzero elements have inverse is called a **division algebra**.*

Example 2.10

In the standard basis $\{\mathbf{e}_i\}_{i=0}^3$ of $\mathbb{R}^4$, choose the structure constants as follows:

$$\mathbf{e}_0^2 = -\mathbf{e}_1^2 = -\mathbf{e}_2^2 = -\mathbf{e}_3^2 = \mathbf{e}_0,$$

$$\mathbf{e}_0\mathbf{e}_i = \mathbf{e}_i\mathbf{e}_0 = \mathbf{e}_i \quad \text{for } i = 1, 2, 3,$$

$$\mathbf{e}_i\mathbf{e}_j = \epsilon_{ijk}\mathbf{e}_k \quad \text{for } i, j, k = 1, 2, 3, i \neq j,$$

where ϵ_{ijk} is completely antisymmetric in all its indices (therefore vanishing if any two of its indices are equal). This algebra is called the **algebra of quaternions** and denoted by $\mathbb{H}$. In this context, $\mathbf{e}_0$ is usually denoted by 1, and $\mathbf{e}_1, \mathbf{e}_2$ and $\mathbf{e}_3$ by i, j , and k , respectively, and one writes $q = x + iy + jz + kw$ for an element of $\mathbb{H}$. It then becomes evident that $\mathbb{H}$ is a generalisation of $\mathbb{C}$. In analogy with $\mathbb{C}$, x is called the **real part** of q , and (y, z, w) the **pure part** of q . Similarly, the **conjugate** of q is $q^* = x - iy - jz - kw$. It is convenient to write $q = x_0 + \mathbf{x}$, where $\mathbf{x}$ is a three-dimensional vector. Then $q^* = x - \mathbf{x}$.

Furthermore, one can show that, with $q = x_0 + \mathbf{x}$ and $p = y_0 + \mathbf{y}$,

$$qp = x_0 y_0 - \mathbf{x} \cdot \mathbf{y} + x_0 \mathbf{y} + y_0 \mathbf{x} + \mathbf{x} \times \mathbf{y}.$$

Changing $\mathbf{x}$ to $-\mathbf{x}$ and $\mathbf{y}$ to $-\mathbf{y}$ in the expression above, one gets

$$q^* p^* = x_0 y_0 - \mathbf{x} \cdot \mathbf{y} - x_0 \mathbf{y} - y_0 \mathbf{x} + \mathbf{x} \times \mathbf{y},$$

which is not equal to $(qp)^*$. However, it is easy to show that $(qp)^* = p^* q^*$. Substituting q^* for p in the previous expression for qp , we get $qq^* = x_0^2 + |\mathbf{x}|^2$. The **modulus** of q , denoted by $|q|$ is (similar to the modulus of a complex number) given by $|q| = \sqrt{qq^*}$. If $q \neq 0$, then $q^*/(x_0^2 + |\mathbf{x}|^2)$ is the inverse of q . Thus the algebra of quaternions is a division algebra.

It is not hard to show that

$$q^n = \sum_{k=0}^{[n]/2} (-1)^k \binom{n}{2k} x_0^{n-2k} |\mathbf{x}|^{2k} + \sum_{k=0}^{[n]/2} (-1)^k \binom{n}{2k+1} x_0^{n-2k-1} |\mathbf{x}|^{2k} \mathbf{x},$$

where $[n] = n$ if n is even and $[n] = n - 1$ if n is odd.

Definition

*Let $\mathcal{A}$ be an associative algebra. A subset $S \subset \mathcal{A}$ is called the **generator** of $\mathcal{A}$ if every element of $\mathcal{A}$ can be expressed as a linear combination of the products of elements in S .*

Definition

Let $\mathcal{A}$ and $\mathcal{B}$ be algebras. A linear map $\phi : \mathcal{A} \rightarrow \mathcal{B}$ is called an **algebra homomorphism** if $\phi(\mathbf{ab}) = \phi(\mathbf{a})\phi(\mathbf{b})$. An injective [respectively, surjective] (respectively, bijective) algebra homomorphism is called, a **monomorphism** [respectively, **epimorphism**] (respectively, **isomorphism**). An isomorphism of an algebra onto itself is called an **automorphism**. The family of homomorphisms from an algebra $\mathcal{A}$ to an algebra $\mathcal{B}$ is denoted by $\text{Hom}(\mathcal{A}, \mathcal{B})$.

Proposition 2.11

Let $\mathcal{A}$ and $\mathcal{B}$ be algebras. Let $\{\mathbf{e}_i\}$ be a basis of $\mathcal{A}$ and $\phi : \mathcal{A} \rightarrow \mathcal{B}$ a linear transformation. Then ϕ is an algebra homomorphism if and only if

$$\phi(\mathbf{e}_i \mathbf{e}_j) = \phi(\mathbf{e}_i) \phi(\mathbf{e}_j).$$

Definition

Let $\mathcal{A}$ and $\mathcal{B}$ be unital algebras. A homomorphism $\phi : \mathcal{A} \rightarrow \mathcal{B}$ is called **unital** if $\phi(\mathbf{1}_{\mathcal{A}}) = \mathbf{1}_{\mathcal{B}}$.

Proposition 2.12

Let $\mathcal{A}$ and $\mathcal{B}$ be unital algebras. If $\phi : \mathcal{A} \rightarrow \mathcal{B}$ is an epimorphism, then ϕ is unital.

Definition

*An endomorphism ω of $\mathcal{V}$ whose square is $\mathbf{1}$ is called an **involution**.*

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Remark

Involutions do not affect the identity of an algebra.

Theorem 2.13

Let $\mathcal{U}$ and $\mathcal{V}$ be two isomorphic linear spaces. Then the algebras $\mathcal{L}(\mathcal{U})$ and $\mathcal{L}(\mathcal{V})$ are isomorphic as algebras.

Definition

A linear space endomorphism $\mathbf{D} : \mathcal{A} \rightarrow \mathcal{A}$ is called a **derivation** on $\mathcal{A}$ if it has the property

$$\mathbf{D}(\mathbf{ab}) = [\mathbf{D}(\mathbf{a})] \mathbf{b} + [\mathbf{D}(\mathbf{b})] \mathbf{a}.$$

Theorem 2.14

Let $\{\mathbf{e}_i\}_{i=1}^N$ be a basis of the algebra $\mathcal{A}$. Then the linear space endomorphism $\mathbf{D} : \mathcal{A} \rightarrow \mathcal{A}$ is a derivation on $\mathcal{A}$ if and only if

$$\mathbf{D}(\mathbf{e}_i \mathbf{e}_j) = \mathbf{D}(\mathbf{e}_i) \cdot \mathbf{e}_j + \mathbf{e}_i \cdot \mathbf{D}(\mathbf{e}_j) \quad \text{for } i, j = 1, 2, \dots, N.$$

Theorem 2.15

*Every derivation $\mathbf{D}$ satisfies the **Leibniz formula***

$$\mathbf{D}^n(\mathbf{ab}) = \sum_{k=0}^n \binom{n}{k} \mathbf{D}^k(\mathbf{a}) \cdot \mathbf{D}^{n-k}(\mathbf{b}).$$

Theorem 2.16

*The set $\mathcal{D}(\mathcal{A})$ of derivations of $\mathcal{A}$ forms an algebra, the **derivation algebra** of $\mathcal{A}$ under the product*

$$\mathbf{D}_1 \bullet \mathbf{D}_2 = \mathbf{D}_1 \mathbf{D}_2 - \mathbf{D}_2 \mathbf{D}_1.$$

Definition

A nonzero element $\mathbf{a} \in \mathcal{A}$ is called **nilpotent** if $\mathbf{a}^k = \mathbf{0}$ for some positive integer k . The smallest such integer is called the **index** of $\mathbf{a}$. A subalgebra $\mathcal{B}$ of $\mathcal{A}$ is called **nil** if all elements of $\mathcal{B}$ are nilpotent. $\mathcal{B}$ is called a **nilpotent of index ν** if $\mathcal{B}^\nu = \{\mathbf{0}\}^\dagger$ and $\mathcal{B}^{\nu-1} \neq \{\mathbf{0}\}$. A nonzero element $\mathbf{P} \in \mathcal{A}$ is called **idempotent** if $\mathbf{P}^2 = \mathbf{P}$.

$^\dagger \mathcal{B}^k$ is the collection of products $\mathbf{a}_1, \mathbf{a}_2 \cdots \mathbf{a}_k$

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Proposition 2.17

The identity element is the only idempotent in a division algebra.

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Theorem 2.18

A nil ideal is nilpotent.

Definition

Let $\mathcal{A}$ be an associative algebra with identity $\mathbf{1}$. For any fixed element $\mathbf{a} \in \mathcal{A}$, consider the set $\mathcal{P}[\mathbf{a}]$ of elements of the algebra of the form

$$p(\mathbf{a}) \equiv \sum_{k=0}^{\infty} \alpha_k \mathbf{a}^k, \quad \alpha_k \in \mathbb{C},$$

in which only a finite number of the terms in the series are nonzero. The set $\mathcal{P}[\mathbf{a}]$ is a commutative algebra with identity called the **polynomial algebra** generated by $\mathbf{a}$. The coefficient of the highest power of $\mathbf{a}$ in $p(\mathbf{a})$ is called the **leading coefficient** of p , and α_0 is called the **scalar term**. A polynomial with leading coefficient 1 is called **monic**. The highest power of $\mathbf{a}$ in p is called the **degree** of p and denoted by $\deg p$. A nonzero polynomial of the form $\alpha_n \mathbf{a}^n$ (no sum) is called a **monomial** of degree n .

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The linear map $\mathbf{d} : \mathcal{P}[\mathbf{a}] \rightarrow \mathcal{P}[\mathbf{a}]$ defined by

$$\mathbf{d}\mathbf{a}^k = \begin{cases} k\mathbf{a}^{k-1} & k \geq 1, \\ 0 & k = 0, \end{cases}$$

is called the **differentiation map** in $\mathcal{P}[\mathbf{a}]$.

Theorem 2.19

The differentiation map $\mathbf{d}$ is a derivation of $\mathcal{P}[\mathbf{a}]$. We denote $\mathbf{d}(p)$ by p' .

Definition

The polynomial $\mathbf{d}^r(p)$ is called the **r th derivative** of p and denoted by $p^{(r)}$. We extend the notation by defining $p^{(0)} = p$. The formula

$$p(\mathbf{a} + \mathbf{b}) = \sum_{r=0}^n \frac{p^{(r)}(\mathbf{a})}{r!} \mathbf{b}$$

is called the **Taylor formula** for p .